CEH-YOLO for Underwater Object Detection

Abstract

This thesis investigates the integration of CEH modules into the YOLOv8 framework for underwater object detection. The work focuses on dataset preparation, annotation conversion, model integration, and pipeline validation. Preliminary results confirm the correctness of the implementation, providing a basis for future large-scale evaluation.

Contents

1 Methodology **1**
1.1 Overall Approach 1
1.2 Baseline Model 1
1.3 CEH Module Integration 1
1.4 Dataset Description 1
1.5 Annotation Conversion 2
1.6 Dataset Structure 2
1.7 Data Validation 2
1.8 Training Strategy 2
1.9 Implementation Environment 2
1.10 Summary 3

1 Methodology

This section describes the implementation and experimental design of the proposed approach. The objective is to integrate CEH modules into the YOLOv8 framework and evaluate their applicability for underwater object detection.

1.1 Overall Approach

The workflow consists of the following stages:

- selection of YOLOv8 as baseline model,
- integration of CEH modules,
- dataset preparation and conversion,
- validation of the pipeline through sanity training.

1.2 Baseline Model

YOLOv8 is used as the baseline object detection model. It follows a one-stage detection paradigm and consists of three main components:

- backbone (feature extraction),
- neck (feature aggregation),
- head (object prediction).

1.3 CEH Module Integration

To improve detection in underwater environments, CEH-based components were integrated into the YOLOv8 architecture:

- High-Order Deformable Attention (HDA),
- Enhanced Spatial Pyramid Pooling Fast (ESPPF).

These modules enhance feature representation and multi-scale detection capability.

1.4 Dataset Description

The dataset consists of underwater images with four object classes:

- echinus,
- starfish,
- holothurian,
- scallop.

The dataset includes training and validation splits.

1.5 Annotation Conversion

The original annotations were provided in COCO format and converted into YOLO format.

Bounding box conversion is defined as:

$$x_{center} = x_{min} + \frac{w}{2}, \quad y_{center} = y_{min} + \frac{h}{2} \tag{1}$$

$$x = \frac{x_{center}}{W}, \quad y = \frac{y_{center}}{H}, \quad w = \frac{w}{W}, \quad h = \frac{h}{H} \tag{2}$$

Final YOLO label format:

```
class x_center y_center width height
```

1.6 Dataset Structure

The dataset was organized as follows:

- images/train
- images/val
- labels/train
- labels/val

A `data.yaml` file defines class names and dataset paths.

1.7 Data Validation

Before training, the dataset was validated by:

- checking label-image consistency,
- verifying coordinate ranges,
- removing corrupt annotations.

1.8 Training Strategy

A sanity training run with one epoch was performed to validate the pipeline. The objective was not performance evaluation but verification of:

- correct data loading,
- proper label parsing,
- successful training execution.

1.9 Implementation Environment

The implementation was conducted using:

- Python,
- Ultralytics YOLOv8 framework,
- local development environment.

1.10 Summary

The methodology establishes a complete pipeline for integrating CEH modules into YOLOv8 and validating the system through structured preprocessing and controlled training.